

ChE 330 – Polymer Science Fall 2022

Learning Outcome 3: An ability to communicate effectively with a range of audiences (written communication)

Rubric Assessment Table

Performance Indicator (PI)	Level 3: Satisfactory	Level 2: Developing	Level 1: Unsatisfactory
1. Articulation of ideas	Articulates ideas clearly and concisely	Articulates ideas, but writing is somewhat disjointed, superfluous or difficult to follow	Text rambles, points made are only understood with repeated reading, and key points are not organized
2. The organization of the written materials	Organizes written materials in a logical sequence to enhance the reader's comprehension (paragraphs, subheading, etc.)	Material are generally organized well, but paragraphs combine multiple thoughts or sections and sub-sections are not identified clearly	Little or no structure or organization; no subheadings or proper paragraph structure used
4. The presentation of the written work	Written work is presented neatly and professionally	Work is not neatly presented throughout	Work is not presented neatly
5. Grammar and spelling	Grammar and spelling are correct	One or two spelling/grammar errors per page	Spelling/grammar errors present throughout more than 1/3 of the paper
6. Writing style	Uses good professional writing style	Style is informal or inappropriate, jargon is used, improper voice, tense...	The writing style is inappropriate for the audience and for the assignment
7. Prescribed format	Conforms to the prescribed format	The prescribed format is only followed in some portions of the paper	The prescribed format is not followed

PI1, PI2, PI4, and PI5 were maximized. Students were provided with templates and given an example to reference. Additionally, the template included guiding questions, which significantly helped students meet the assignment requirements.

However, student performance on PI6 and PI7 was minimal. These outcomes require the use of ACS reference style, which is housed in an external database. This citation style is likely new to students, who have primarily been trained in MLA or APA formats through composition courses. While citation generators are available, many are not free, so students likely completed references manually.

Students also demonstrated difficulty setting margins to 1 inch, indicating a gap in basic Microsoft Word skills. In the future, this could be addressed by including a brief tutorial or providing resources for review. However, at this stage in their academic journey, students should have already been exposed to these formatting skills.

Thermal processing of plastics from waste electrical and electronic equipment for hydrogen production

<https://doi.org/10.1016/j.jaap.2012.09.014>

Researchers collected WEEE plastic waste from waste recycled electronic equipment produced from commercial recycling. Glass and circuit boards are removed from electronic products and recycled. The purpose of these recycling components is to be used in the production of plastics ABS and HIPS (mainly ABS). In this experiment, a 1kg sample of WEEE plastics was obtained and an elemental analysis of this sample was carried out. These plastics are used to try and produce a higher hydrogen yield. This experiment is completed by ABS being processed in a two-stage gasification reaction system to produce hydrogen. In this experiment ABS goes through a first reactor where it is pyrolyzed then goes through a second reactor where heat is added along with a catalyst. ABS was one of the main components of the plastic waste reacted in the system. The volatile products obtained after the gasification process were passed through two condensers to cool the plastics. Reaction reproducibility was tested and repeated to ensure research results accuracy. ABS showed an increase of 15.4 wt% and 16.2 wt% from the 5 wt% and 10 wt% nickel catalysts. Results showed that introduction of steam and a catalyst to the two-stage gasification reaction of these plastics increases the yield of Hydrogen. Since the wt% of gas yield increased from using ABS plastics throughout the process. This process showed great promise in the production of hydrogen. This helped me understand that gasification reactions using ABS can aid strongly in the production of Hydrogen. To conclude, ABS has many applications and uses, production of larger amounts of hydrogen is one of the most vital uses of ABS. This leads to production of electricity, industry powering, and will make a huge difference on carbon emissions.

Acomb, J. C.; Nahil, M. A.; Williams, P. T. Thermal processing of plastics from waste electrical and electronic equipment for hydrogen production. *Journal of Analytical and Applied Pyrolysis* 2013, 103, 320-327. DOI: 10.1016/j.jaap.2012.09.014.